



Finacle CI/CD C++ Project

Release Notes

Project Details

Project Name	Finacle C++ CI/CD
Project Type	Software Development
Project Version	1.0
Name of Releaser	Kennedy N. Gatimu
File Mode of Delivery	Pipeline
Prepared By	Kennedy N. Gatimu

Background

The Finacle Auto Deployment Project was developed to address and mitigate the problems associated with manual script deployment. The key issues with manual deployments include:

- Lack of proper pre-deployment approvals.
- Lack of proper version control.
- Inability to perform continuous integration and deployment.
- Scripts prone to accidental deletion.
- Lack of a central repository outside Finacle servers.
- Peer collaboration challenges.

Solution Overview

This release introduces a new automated deployment process for on-premises Finacle deployments using Azure DevOps pipelines. This streamlines the process, improves governance, and fosters collaboration through:

- **Centralized Repository:** Scripts are securely stored in a central Azure Repos git repository.
- **Version Control:** Git ensures proper version control, enabling rollbacks and historical tracking.
- **Pre-Deployment Approvals:** An approval stage in the deployment pipeline enforces authorization before deployment.
- **Reduced Server Access:** Script execution utilizes a dedicated service principal, minimizing write privileges on Finacle servers.

- **Enhanced Collaboration:** Git facilitates team collaboration and code sharing.

New Features

This release introduces a two-stage CI/CD pipeline for automated deployments:

Continuous Integration (CI) Pipeline

- Triggers on commits to the main(production) branch or release branch of the Azure Repos containing deployment scripts.
- Automated build and validation of script files stored in the Azure repository.
- Packages the deployment scripts for release.

Continuous Delivery (CD) Pipeline

- Triggers on successful completion of the CI pipeline or upon manual initiation.
- Automated deployment of scripts to on-premises environments.
- Environment-specific configurations to ensure seamless deployment across various stages (dev, uat, prod).
- Performs post-deployment tasks (e.g., automatic clean ups).

Azure Repository

- Centralized storage for all script files.
- Branching strategy implemented to manage feature development and releases.
- Access control and permissions set up to ensure security and integrity of the repository.

Enhancements

- Implement comprehensive logging throughout the CI/CD pipelines, capturing script execution details, environment variables, and success/failure messages.
- Implement robust error handling within the pipelines to gracefully handle script failures and unexpected errors.
- Implement performance monitoring for the CI/CD pipelines themselves, tracking execution times and identifying bottlenecks for optimization.

Benefits

By implementing CI/CD pipelines:

- Reduced manual effort and error-prone deployments.
- Improved consistency and repeatability through automation.
- Faster deployments with quicker feedback loops.
- Increased traceability through pipeline logs for troubleshooting.
- Improved collaboration.

This translates to reduced costs, faster deployment time, and a more robust Finacle environment.